

E.3 / PAPER 1 / QUESTION 45

29 April 2025 / TZ2 (file label) / HL Paper 1A / original Q38 / 2225-6713

45. A pure sample of 4.4 g of Radon-222 has an activity of 2.5×10^{16} Bq.

What is the decay constant of Radon-222?

- A. $2.1 \times 10^{-10} \text{ s}^{-1}$
- B. $8.3 \times 10^{-10} \text{ s}^{-1}$
- C. $2.1 \times 10^{-6} \text{ s}^{-1}$
- D. $8.3 \times 10^{-6} \text{ s}^{-1}$